4.7 Transformations of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.7 · Student sheet

I can…

- describe transformations of polynomial functions.
- write rules for transformed polynomial functions.
- graph transformed polynomial functions.

Vocabulary · fill in the missing word

Horizontal translation — _____

Vertical translation — _____

Reflections — −f(x) in the _____; f(−x) in the y-axis

Stretches and shrinks — a · f(x) vertically; f(ax) _____

Part 1 · Match-up

Descriptions: A. 5 units down from x^4 B. horizontal shrink of x^3 by ½ C. x^4 moved 1 right and 2 up D. 3 units left of x^3 E. reflection of x^3 in the x-axis F. vertical shrink of x^4 by ⅓

Function	Description
$(x + 3)^3$	
$x^4 - 5$	
$-x^3$	
$(2x)^3$	
$\frac{1}{3}x^4$	
$(x - 1)^4 + 2$	

Part 2 · Notes: fill in the blanks

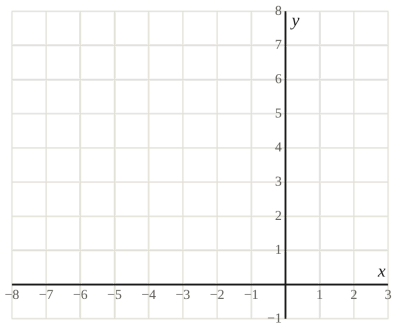
f(x − h) moves the graph _____; f(x) + k moves it _____.

f(−x) reflects in the _____; −f(x) reflects in the _____.

Part 3 · Examples (work them with the class)

Example 1

Describe the transformation of $f(x) = x^4$ represented by $g(x) = (x + 5)^4 + 1$.



Example 2

Describe the transformation of $f(x) = x^3$ represented by $g(x) = -\frac{1}{2}(x - 2)^3$.

Example 3

Let $f(x) = x^3 + x^2 + 3$. Write a rule for g, a reflection of f in the y-axis.

Example 4

Let $f(x) = x^3 - 6$. Write a rule for g, a vertical stretch by a factor of 2 followed by a translation 3 units up.

Part 4 · Partner practice

Partner 1

Describe the transformation of $f(x) = x^3$ represented by $g(x) = (x - 4)^3 - 7$.

Partner 2

Let $f(x) = x^4 - 2x$. Write a rule for g, a horizontal shrink by a factor of $\frac{1}{2}$.